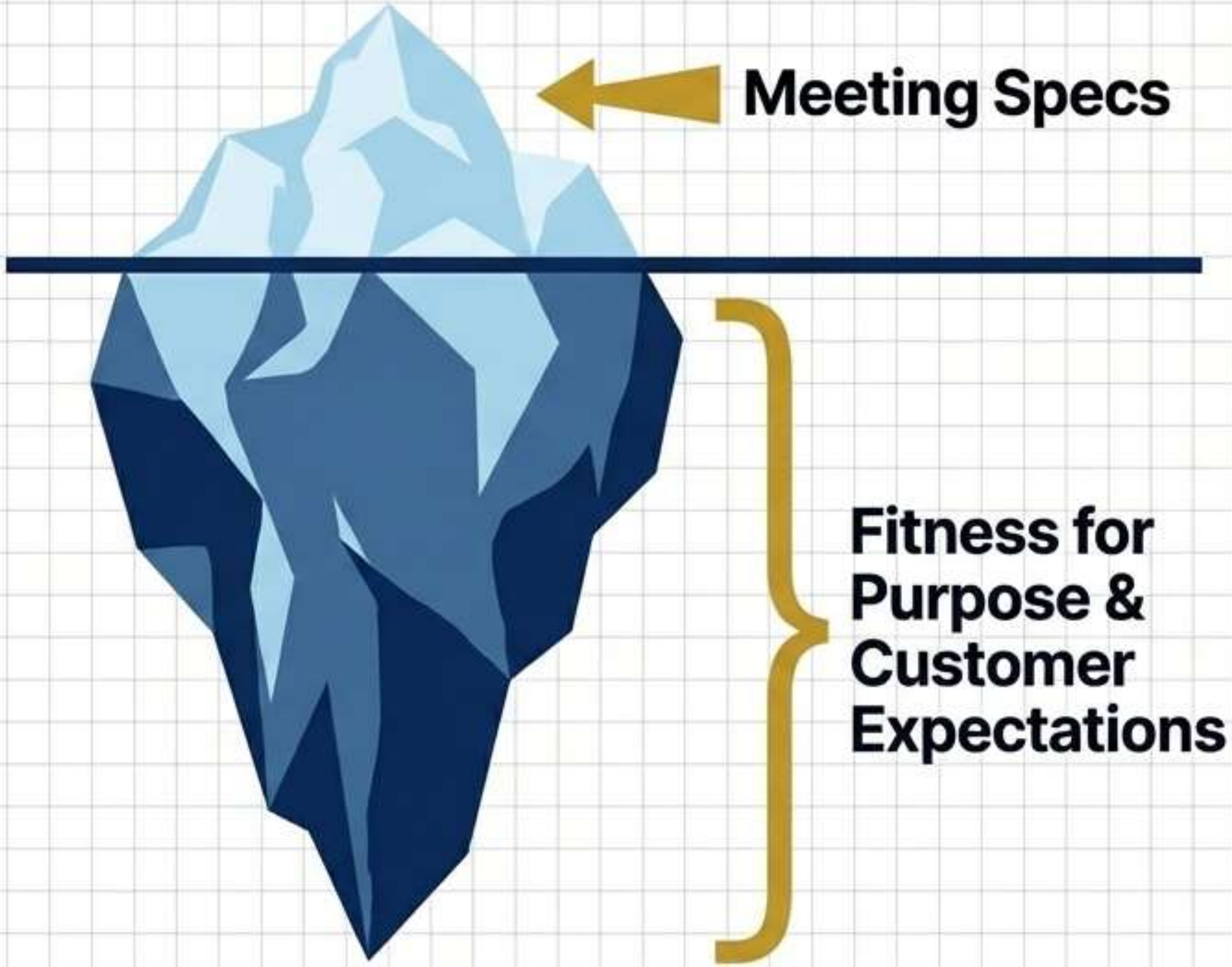
Topic 6: Project Quality Management

IT Project Management (INS3044)

Learning Objectives

- Understand project quality concepts.
- Define quality planning processes.
- Differentiate assurance and control.
- Apply quality management tools.
- Analyze real-world quality failures.

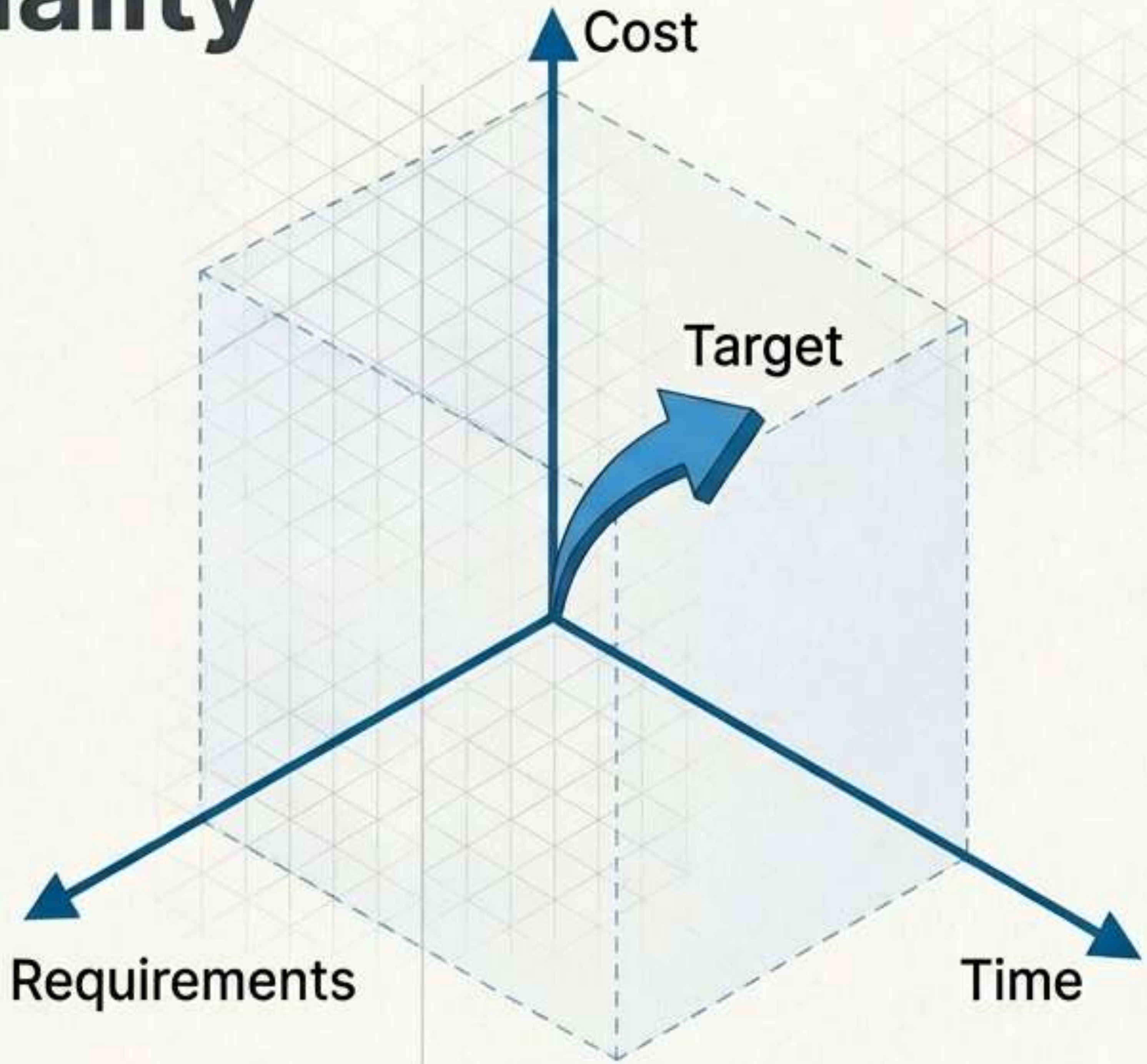
The Concept of Quality



- Meets agreed project specifications.
- Fits intended product purpose.
- Fulfills unstated customer expectations.
- Prevents defects before occurring.
- Requires entire team effort.

Defining Project Quality

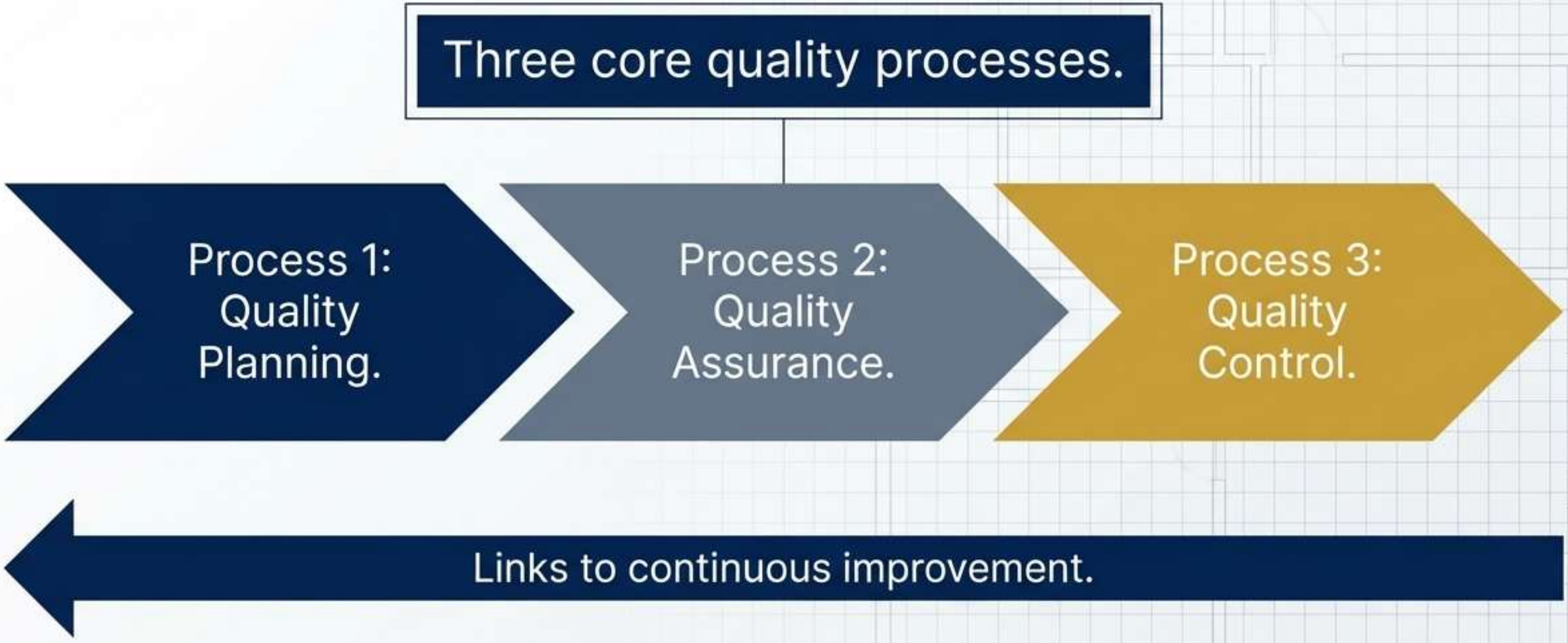
- Meet exact customer specifications.
- Ensure fitness for purpose.
- Eliminate critical project defects.
- Balance time, cost, performance.
- Achieve "good enough" standard.
- Delight the final customer.



The 'Good Enough' Quality Threshold

In IT Project Management, the goal is rarely absolute perfection. We aim for the threshold where critical requirements are met before costs become infinite.

Quality Management Framework



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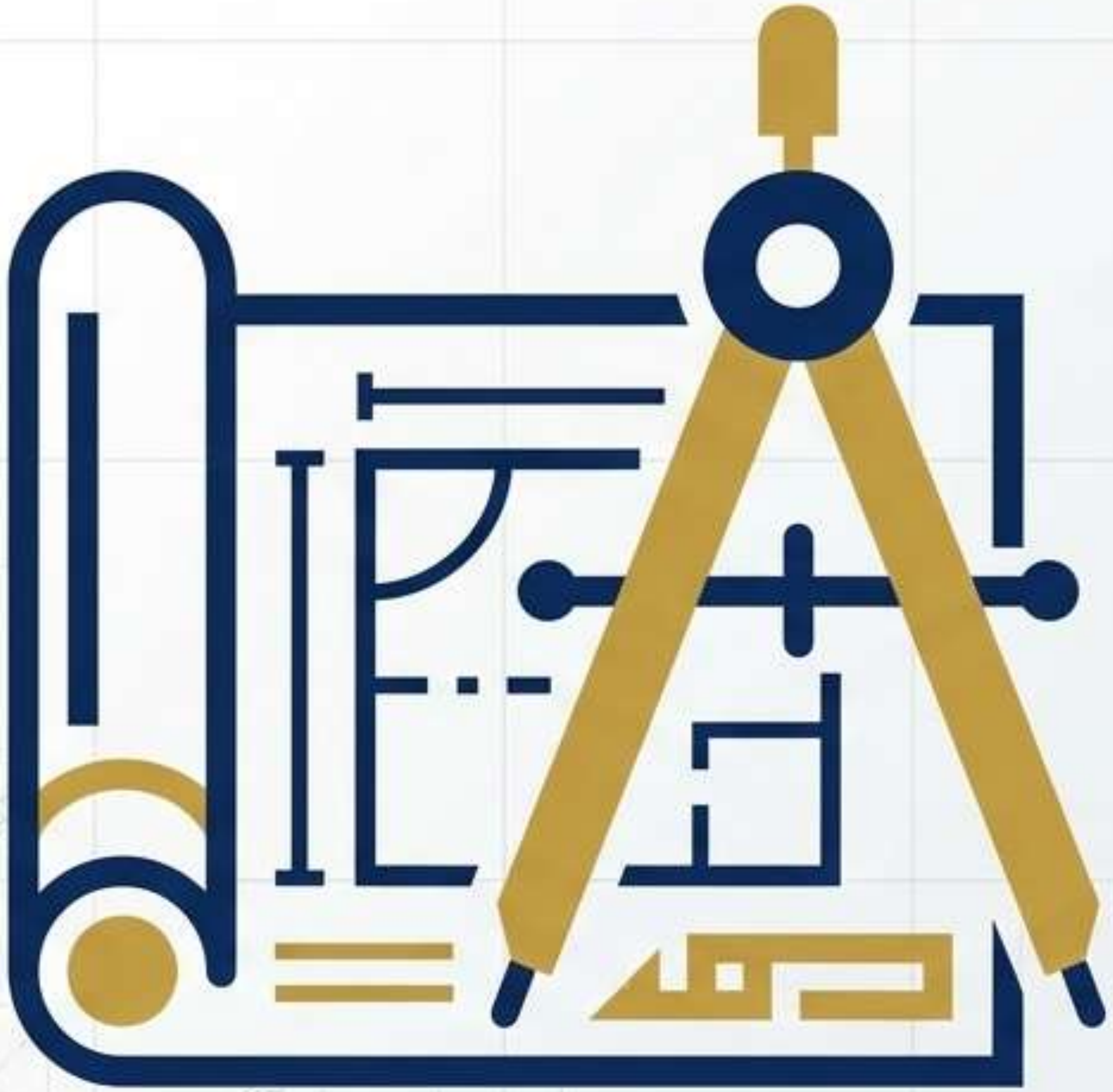
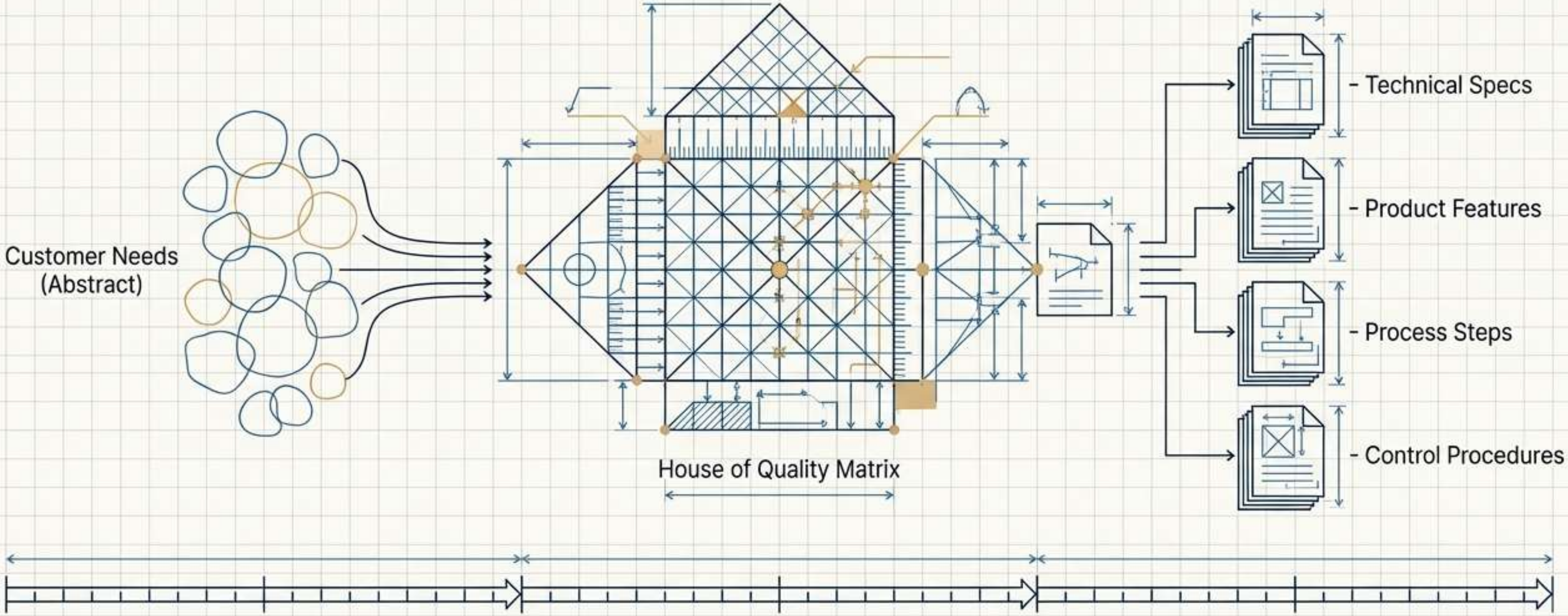
Process 1 - Quality Planning



- Establishes future quality activities.
- Sets requirements and standards.
- Defines organizational policies used.
- Schedules specific quality tasks.
- Assigns clear task responsibilities.

The 'House of Quality' Translation Engine

Quality Function Deployment (QFD) ensures no requirement is lost in translation.



Step 1: Capture market needs.

Step 2: Translate needs into hard engineering measures.

Step 3: Track every activity back to a specific customer requirement (Traceability).

Process 2 - Quality Assurance (QA)

- Executes the quality plan.
- Reduces risk of failing.
- Ensures correct processes used.
- Builds confidence for customer.
- Drives continuous organizational improvement.

QA is the 'Healthy Lifestyle'

Quality Assurance is proactive. We build a healthy system to prevent failures before they occur.

Quality Assurance

Healthy Lifestyle (Prevention)

Building structural processes and systemic habits to ensure things are engineered right the first time.

Quality Control

Medicine (Intervention)

Treating the symptoms, running diagnostics, and applying fixes to eliminate existing nonconformities.

The Quality Assurance Toolkit

QA managers rely on specific proactive processes to ensure the project meets its quality standards during development.

Configuration Management

Tracking design information and maintaining strict version control across all teams.

Design Reviews

Cross-functional evaluation of product viability, manufacturability, and omissions.

Audits

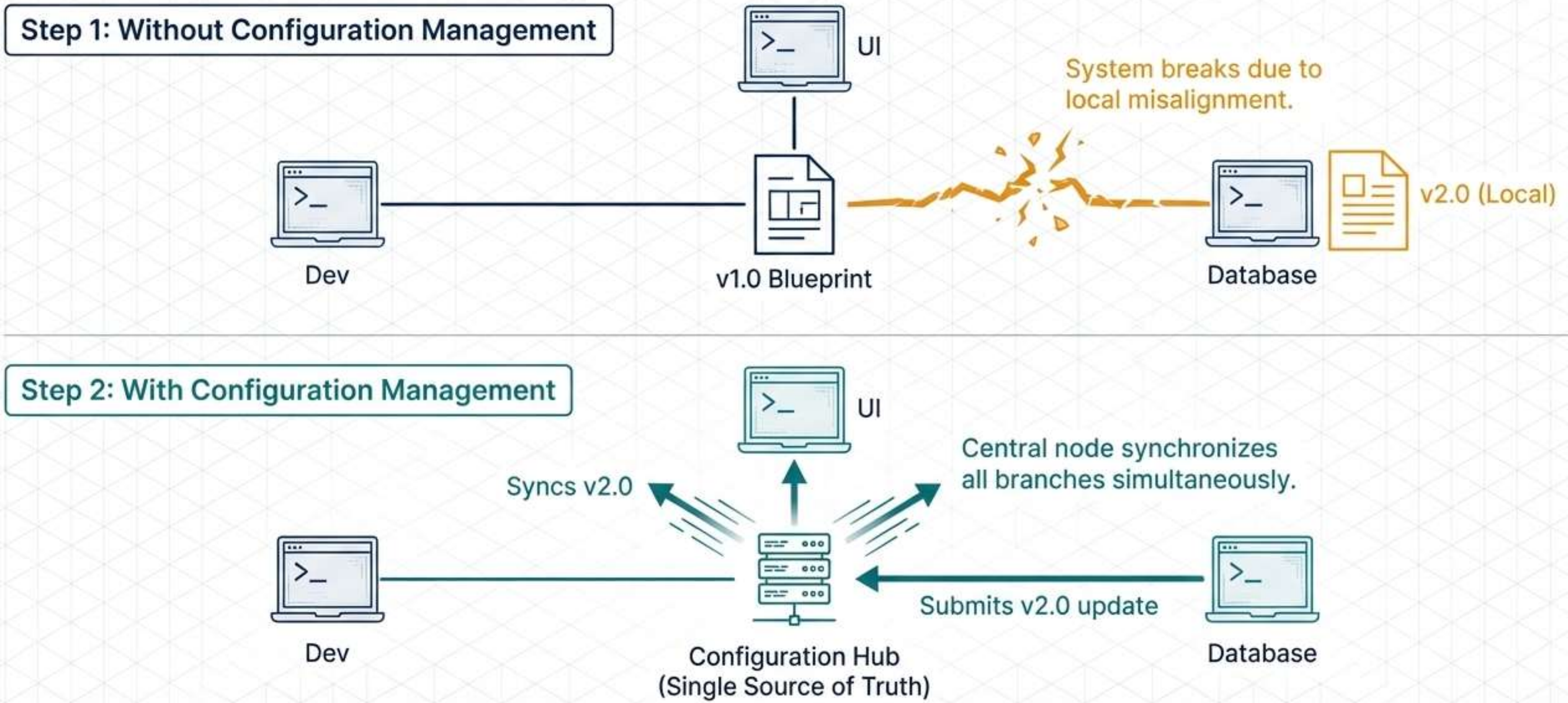
Independent verification of strict compliance with prescribed management processes.

FMEA

Mathematically calculating and prioritizing system risks and potential failures.

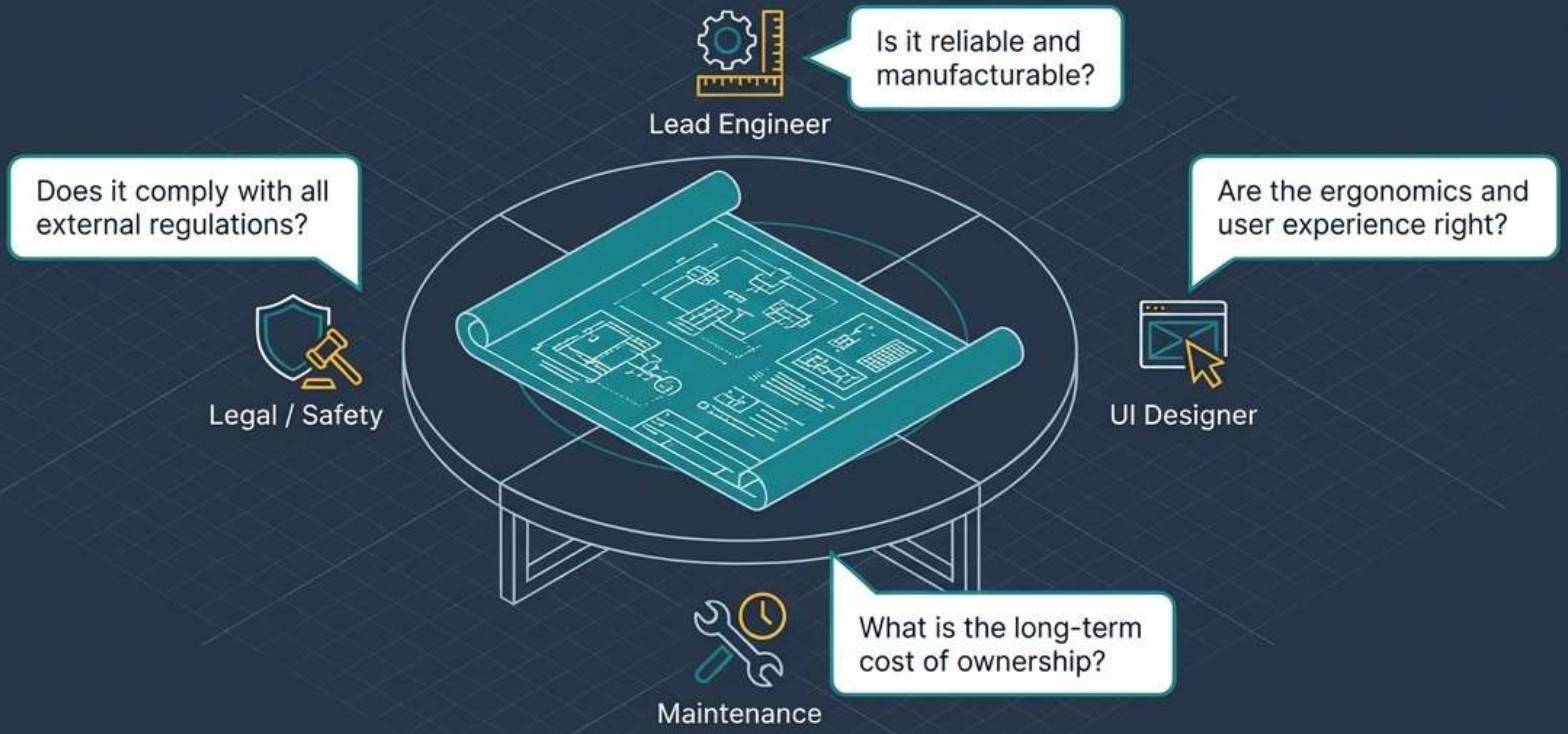
Configuration Management

Ensuring everyone builds from the exact same blueprint.



Design Reviews

A formal, cross-functional evaluation to identify omissions, verify compliance, and ensure manufacturability before building begins.



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Quality Audits

While design reviews check the product, audits check the process.

Note: Audits must be conducted by credible, unbiased internal staff or external third parties to maintain process integrity.

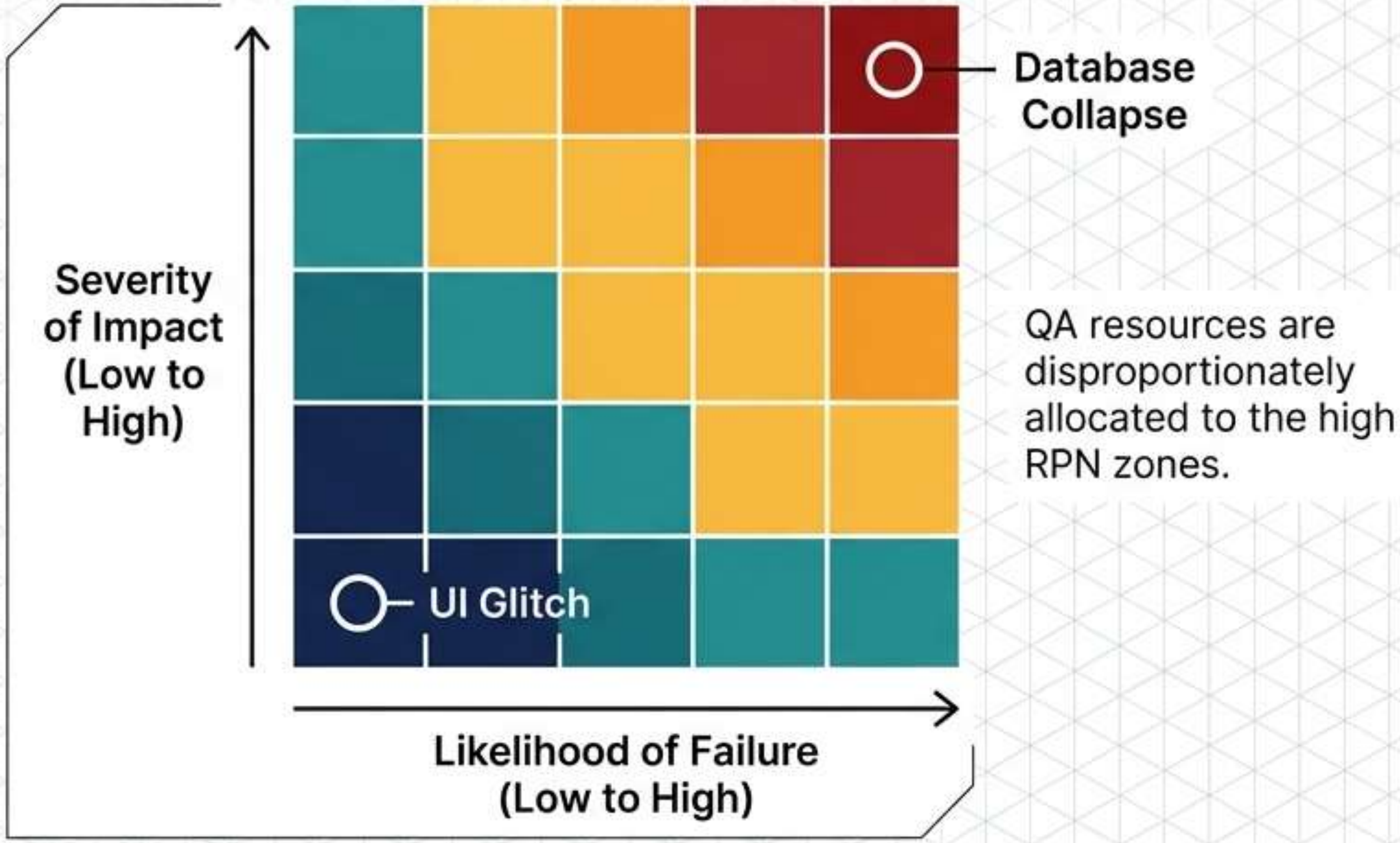
Clarifying the Tools: Reviews vs. Audits

	Design Reviews	Quality Audits
Primary Focus	The technical product and its viability.	The management processes and compliance.
Who is involved?	Cross-functional team (Engineers, Users, Subject Matter Experts).	Independent, unbiased internal staff or external third parties.
Primary Goal	Catching technical omissions and improving manufacturability.	Verifying strict adherence to prescribed procedures.

Prioritizing Preventative Care (FMEA)

QA teams use Failure Mode and Effect Analysis to mathematically calculate which system components need the most attention.

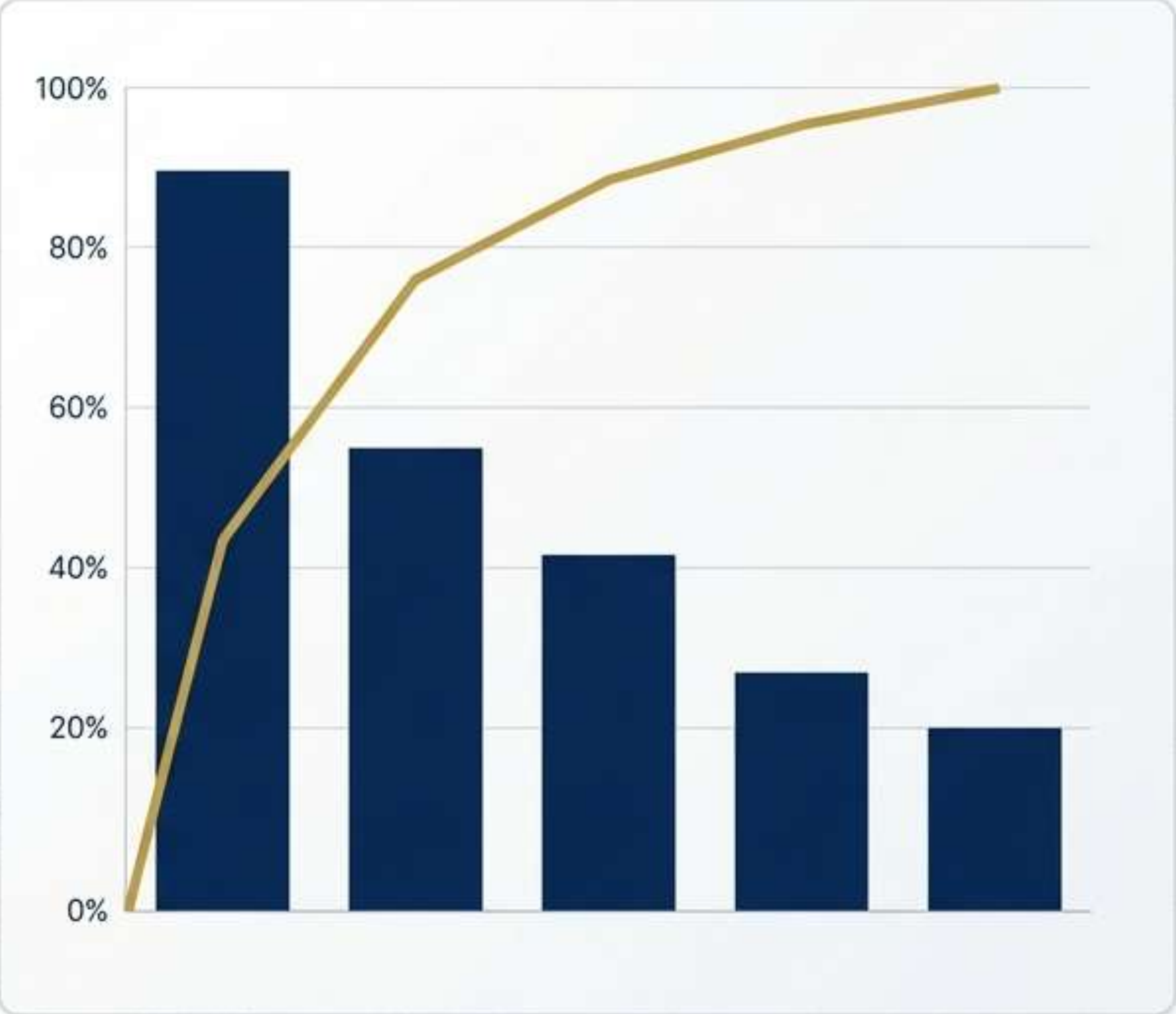
Severity x Likelihood x Detection = Risk Priority Number (RPN)



Process 3 - Quality Control (QC)

- Monitors specific project results.
- Verifies compliance with specifications.
- Identifies existing product defects.
- Requires corrective action implementation.
- Integrates with overall control.

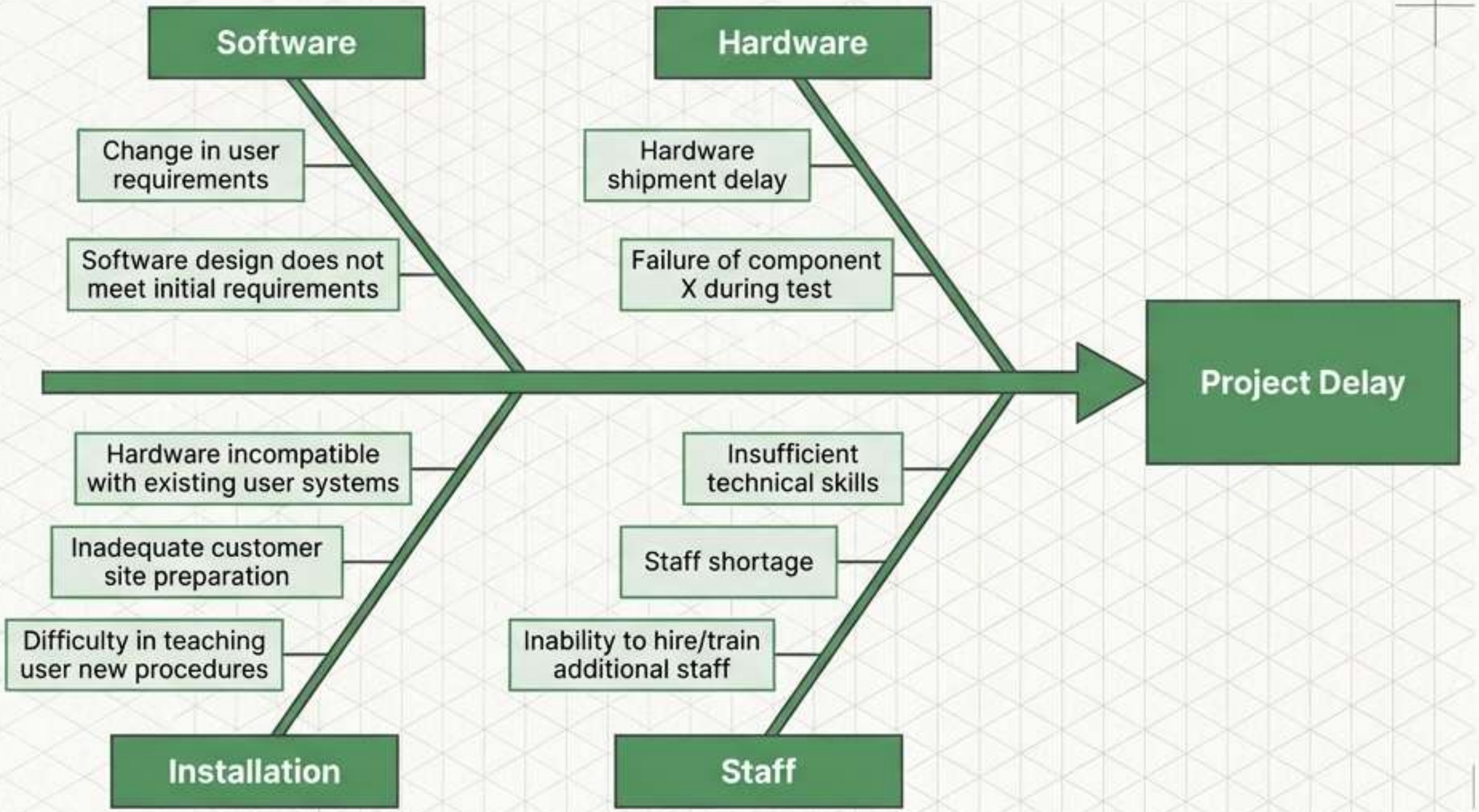
Tools for Quality Control



- Inspection of final deliverables.
- Acceptance testing by customers.
- Pareto diagrams isolate causes.
- Cause-and-effect traces roots.
- Ad-hoc problem solving methods.

QC Tool: Fishbone Diagram

- Identify root problem causes.
- Categorize potential defect sources.
- Analyze hardware and software.
- Examine staff and funds.
- Solve complex quality issues.



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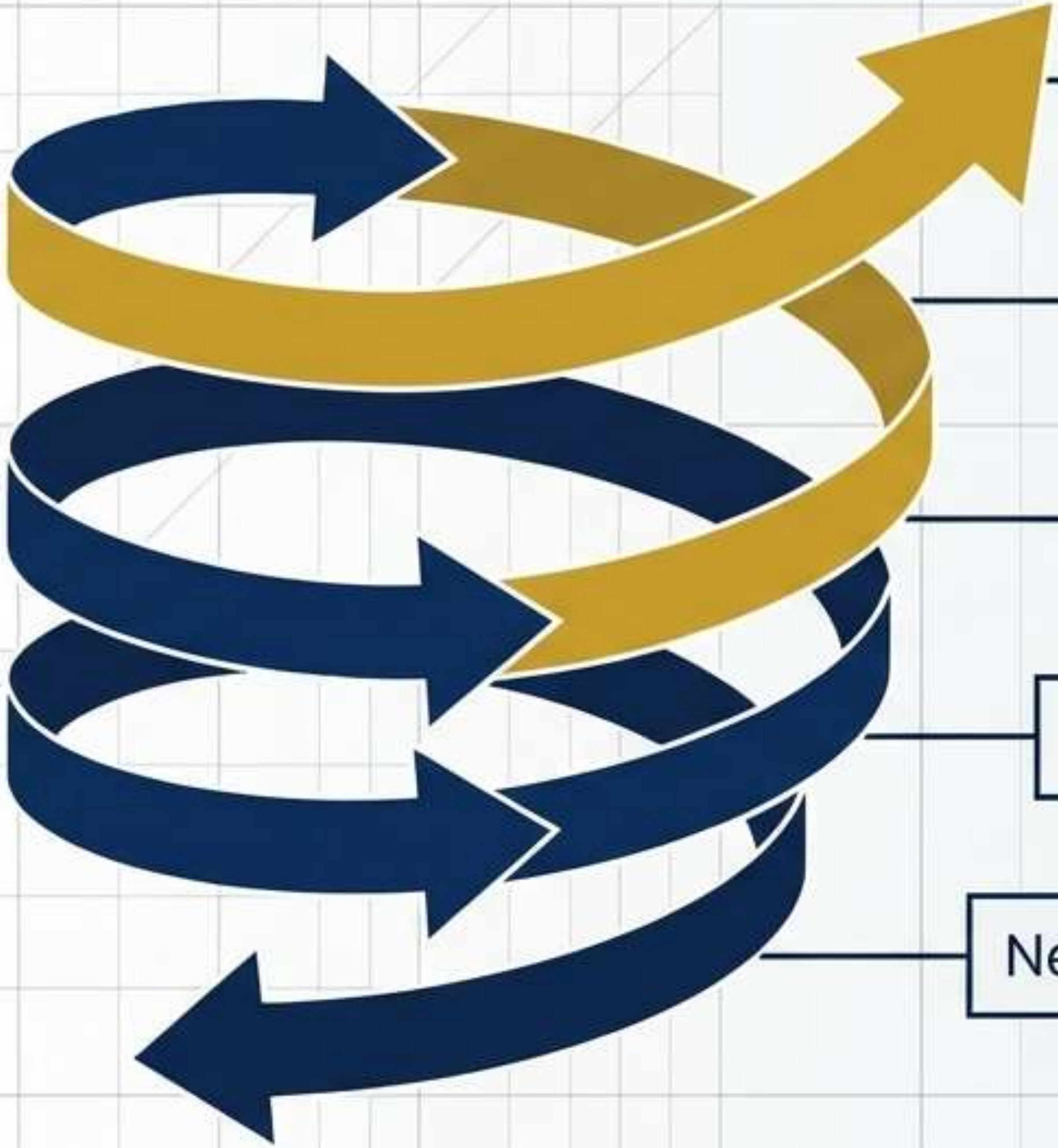
SYNTHESIS: QA vs. QC Matrix

Quality Assurance (QA)	Quality Control (QC)
QA focuses on process.	QC focuses on product.
QA aims to prevent.	QC aims to detect.
QA is proactive management.	QC is reactive management.

Continuous Improvement



Foundation of modern progress.

Learn from every phase.

Conduct post-completion project reviews.

Document all lessons learned.

Never repeat same mistakes.

Case Study: Healthcare.gov Launch Failure

- Website crashed immediately upon launch.
- Users unable to enroll in plans.
- Testing was severely compressed and insufficient.
- Severe performance and security issues.
- Major financial and reputational costs.

Key Takeaways

- Quality planned, not inspected.
- QA prevents; QC detects.
- Focus on fitness purpose.
- Tools must match needs.
- Continuous improvement always required.